

ON THE BASIS OF MATHEMATICS

Christopher Robinson Tompkins III, Independent, Richmond, VA

Corresponding author. Email: chtompki@apache.org,

chtompki@vt.edu, tompkinscr@vcu.edu

We explore and explain the fundamentals of mathematical language(s). Simply.

(0)

*Dedicated to the memory of those killed at Virginia Tech,
and in solidarity with those who were injured, their families,
and the Virginia Tech community.*

*I would like to acknowledge the following people that helped me along the way: mom, dad, Margo,
Nora, Griff Elder (and the 2007 VT Mathematics department and all my friends from there),
George McNulty (and the 2009 USC Mathematics department and all my friends from there),
Andy Lewis (and the VCU 2026 Mathematics Department and all my friends from there), and
those at NASA whom I've worked with.*

*Note. The examples and the idea about using photos for self reference as opposed to using the
drawn hand came from ChatGPT. THE IDEAS!!!*

(1)

MATHS

We have a considerable number of open problems with regards to mathematics. Though there is one above the rest that confuses us. It is simple, it's been used and referred to numerous times. The great antinomy:

$$P : \neg P \quad \text{or} \quad \text{"THIS SENTENCE IS FALSE. . ."}$$

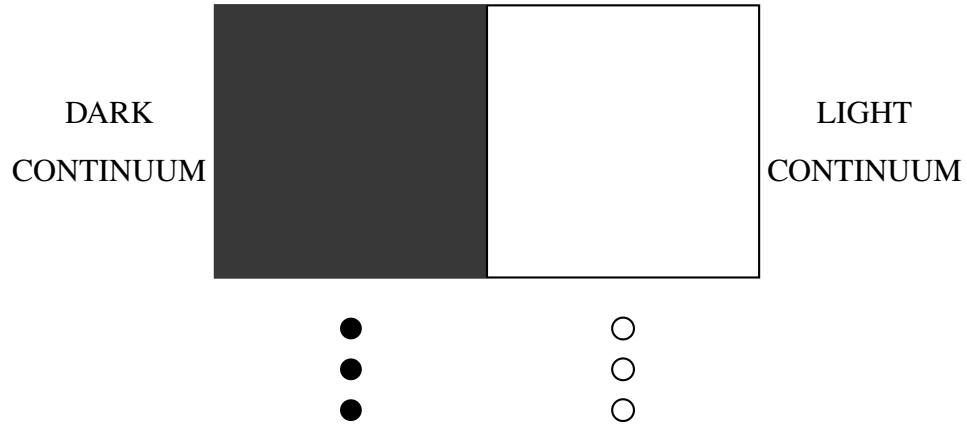
Yes considerable work has been done to "RECTIFY" or FIX this seemingly fundamental flaw in logic. But, in 1934 through his numbering system and Turing's machine, Kurt Gödel showed this sentence to be TRUE.

Now we have a slightly different opinion of the matter, it is the beginning of the language (NATURAL or MECHANICAL) and its collection of ideas. We ask is P TRUE or FALSE, but that's not the answer at all – because it is both TRUE AND FALSE simultaneously and this is good because it gives us the concepts of

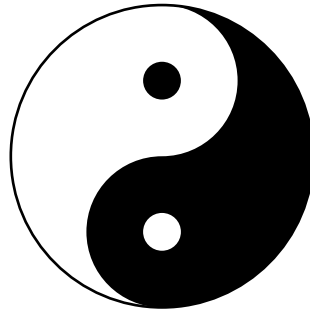
TRUE, FALSE, AND, OR, NOT

implicitly. Yet we're missing something, even though we have TRUTH defined, what is it? A SYSTEM OF REFERENTIAL CAPACITY, so let's define one.

So we begin with the idea of two bounded continuums



OR perhaps



WE CHOOSE DARK AND LIGHT TO DIFFERENTIATE OUR CONTINUUMS OUT OF NATURAL CONVENIENCE. WE THEN ALLOW “DOTS” OR “ATOMS” (IF WE WANT TO GET TO RUSSELL & TARSKI) TO BE PULLED OR CREATED FROM EITHER OF THE CONTINUUM. AN ATOM OR DOT WE TAKE TO BE A CONTIGUOUS PIECE OF CONTINUUM, JUST RELATIVELY SMALLER.

Some quick examples of the power for referential capacity follow as,

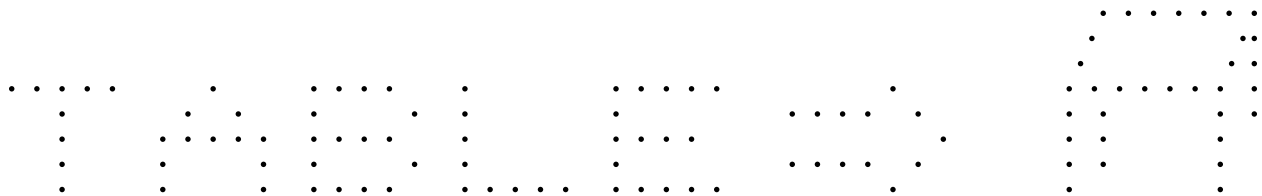
(2)

Once we have the capacity to make characters and references, we can naturally build up our logical connectives:

$$\wedge, \vee, \neg, \rightarrow, \leftrightarrow, \equiv, \exists, \forall, 0, 1, S(n) = n + 1, \{x_0, \dots, x_n\}, \{c_0, \dots, c_n\}.$$

●	●	●	●	●
●				
●	●	●	●	●
				●
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Example 1: Description of the drawing.



Example 2: Dots forming the word “TABLE,” followed by an arrow and a representation of a table.

So, this sufficiently builds our language of discourse, call it L , and we start by adding $P : \neg P$ to the language as the first sentence to bootstrap the definitions of TRUE, FALSE, and our logical connectives.

DEFN IF A STATEMENT CAN BE BUILT FROM DOTS OR ATOMS, WE CALL IT WRITABLE.

Now WRITABILITY is a fundamental property that both RUSSELL AND TARSKI missed in their definitions of the instantiating mechanics of mathematics. IT’S KINDA LIKE MISSING ZERO FOR ALL THOSE YEARS.

QUESTION? DO THERE EXIST IDEAS THAT ARE MORE COMPLEX THAN WRITABLE?

→ YES/NO

THEY MAY EXIST IN OUR MINDS, BUT FOR ALL INTENTS AND PURPOSES, IF WE CANNOT CONVEY THE IDEA THEN IT’S POINT IS MOOT. SO

WRITABILITY IS THE BOUND ON MATHEMATICAL LANGUAGE

THIS EFFECTIVELY IMPLIES THAT THERE IS NO META-LANGUAGE, THERE IS ONLY
THE LANGUAGE OF DISCOURSE

$$L \subseteq M_L \quad \text{BUT} \quad L \neq M_L.$$

$$\text{BUT NOW} \quad L = M_L$$

SO GÖDEL'S COMPLETENESS THEOREM
HOLDS

CURIOUS! COMPLETENESS & INCOMPLETENESS MUST
SIMULTANEOUSLY HOLD OVER L AND M_L
BY THE EXISTENCE OF THE LIAR'S PARADOX,
YET THE LANGUAGE DEFINES THE LANGUAGE?

HILBERT 2. YES & NO

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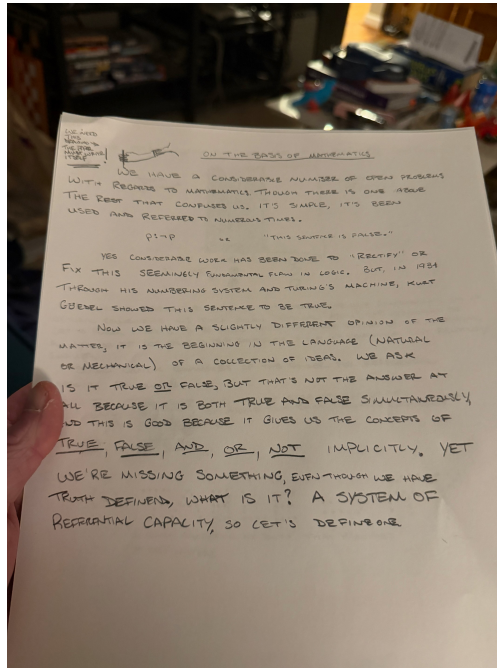


Figure 1: A photo of page 1 of this handwritten paper for the purposes of self referentiality

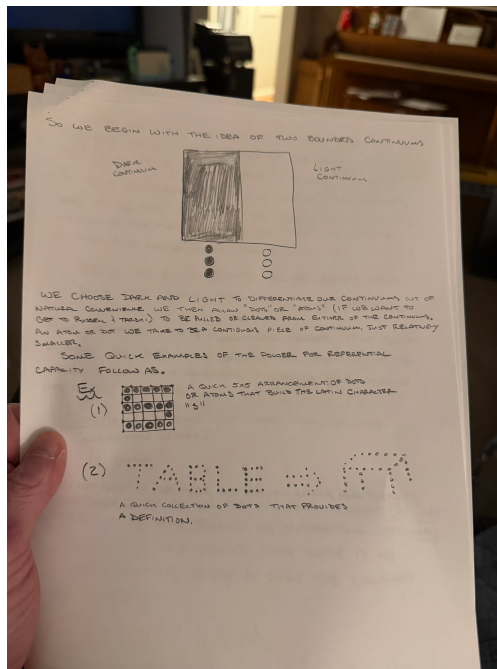


Figure 2: A photo of page 2 of this handwritten paper for the purposes of self referentiality

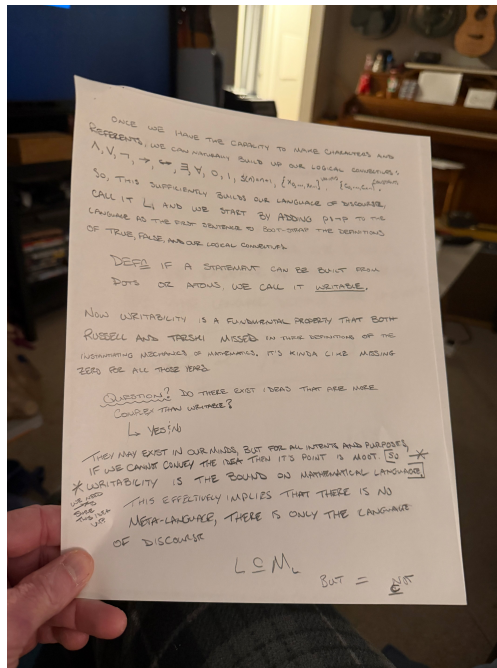


Figure 3: A photo of page 3 of this handwritten paper for the purposes of self referentiality

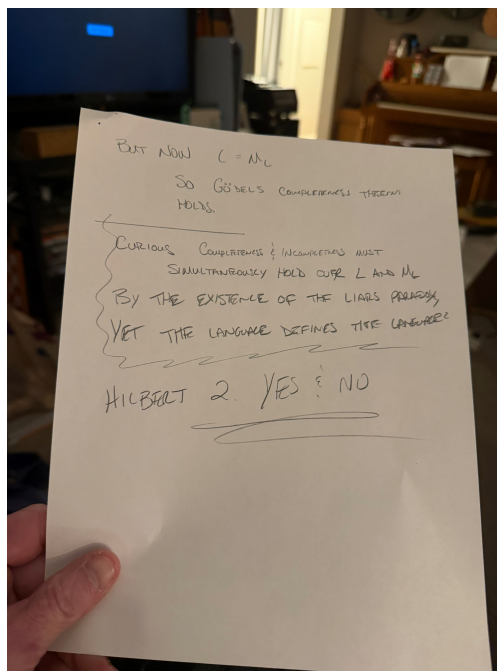


Figure 4: A photo of page 4 of this handwritten paper for the purposes of self referentiality